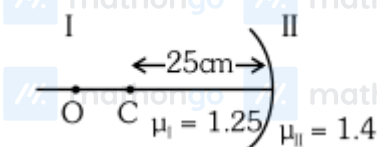


Q1 (20 July 2021 Shift 1)

Region I and II are separated by a spherical surface of radius 25 cm. An object is kept in region I at a distance of 40 cm from the surface. The distance of the image from the surface is :



(1) 55.44 cm

(2) 9.52 cm

(3) 18.23 cm

(4) 37.58 cm

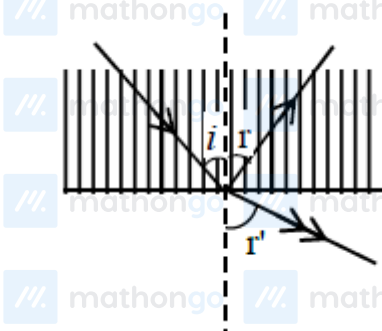
Q2 (20 July 2021 Shift 1)

An object viewed from a near point distance of 25 cm, using a microscopic lens with magnification '6', gives an unresolved image. A resolved image is observed at infinite distance with a total magnification double the earlier using an eyepiece along with the given lens and a tube of length 0.6 m, if the focal length of the eyepiece is equal to ____ cm.

Q3 (22 July 2021 Shift 1)

A ray of light passes from a denser medium to a rarer medium at an angle of incidence i . The reflected and refracted rays make an angle of 90° with each other.

The angle of reflection and refraction are respectively r and r' . The critical angle is given by :



Questions with Answer Keys

MathonGo

(1) $\sin^{-1}(\cot r)$

(2) $\tan^{-1}(\sin i)$

(3) $\sin^{-1}(\tan r')$

(4) $\sin^{-1}(\tan r)$

Q4 (22 July 2021 Shift 1)

A ray of light passing through a prism ($\mu = \sqrt{3}$)

suffers minimum deviation. It is found that the

angle of incidence is double the angle of refraction

within the prism. Then, the angle of prism is _____ (in degrees)

Q5 (25 July 2021 Shift 1)

A ray of laser of a wavelength 630 nm is incident

at an angle of 30° at the diamond-air interface. It is

going from diamond to air. The refractive index of

diamond is 2.42 and that of air is 1. Choose the

correct option.

(1) angle of refraction is 24.41°

(2) angle of refraction is 30°

(3) refraction is not possible

(4) angle of refraction is 53.4°

Q6 (25 July 2021 Shift 2)

A prism of refractive index μ and angle of prism A is placed in the position of minimum angle of deviation. If minimum angle of deviation is also A, then in terms of refractive index

(1) $2 \cos^{-1}\left(\frac{\mu}{2}\right)$

(2) $\sin^{-1}\left(\frac{\mu}{2}\right)$

Questions with Answer Keys

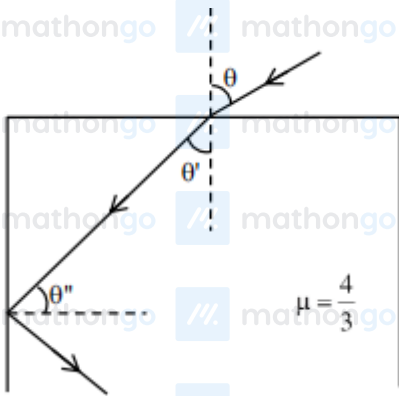
MathonGo

$$(3) \sin^{-1} \left(\sqrt{\frac{\mu-1}{2}} \right)$$

$$(4) \cos^{-1} \left(\frac{\mu}{2} \right)$$

Q7 (25 July 2021 Shift 2)

A ray of light entering from air into a denser medium of refractive index $\frac{4}{3}$, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown. The maximum value of angle θ should be equal to :



$$(1) \sin^{-1} \frac{\sqrt{7}}{3}$$

$$(2) \sin^{-1} \frac{\sqrt{5}}{4}$$

$$(3) \sin^{-1} \frac{\sqrt{7}}{4}$$

$$(4) \sin^{-1} \frac{\sqrt{5}}{3}$$

Q8 (27 July 2021 Shift 1)

A prism of refractive index n_1 and another prism of refractive index n_2 are stuck together (as shown in the figure). n_1 and n_2 depend on λ , the wavelength of light, according to the relation

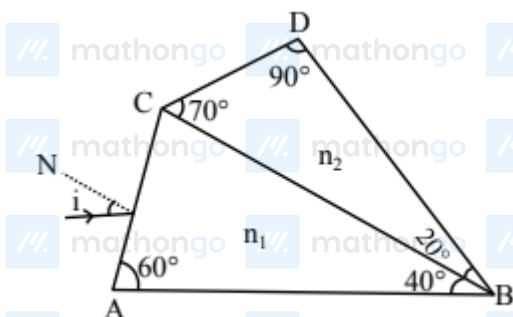
$$n_1 = 1.2 + \frac{10.8 \times 10^{-14}}{\lambda^2} \text{ and } n_2 = 1.45 + \frac{1.8 \times 10^{-14}}{\lambda^2}$$

The wavelength for which rays incident at any angle on the interface BC pass through without bending at that

Questions with Answer Keys

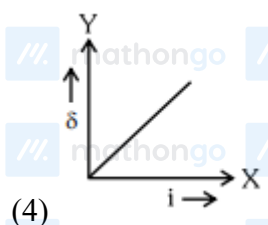
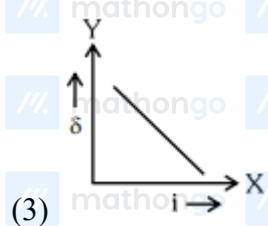
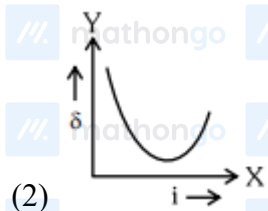
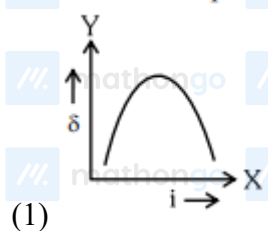
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interface will be _____ nm



Q9 (27 July 2021 Shift 2)

The expected graphical representation of the variation of angle of deviation ' δ ' with angle of incidence ' i ' in a prism is :



Q4 (60)

Q8 (600)

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Hints and Solutions

MathonGo

Q1

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.4}{v} - \frac{1.25}{-40} = \frac{1.4 - 1.25}{-25}$$

$$\frac{1.4}{v} = -\frac{0.15}{25} - \frac{1.25}{40}$$

$$v = -37.58 \text{ cm}$$

Hence option (4)

Q2

For simple microscope,

$$m = 1 + \frac{D}{f_0}$$

$$6 = 1 + \frac{D}{f_0}$$

$$5 = \frac{25}{f_0}$$

$$f_0 = 5 \text{ cm}$$

For compound microscope,

$$m = \frac{\ell \cdot D}{f_0 \cdot f_e}$$

$$12 = \frac{60 \times 25}{5 \cdot f_e}$$

$$f_e = 25 \text{ cm}$$

Q3

$$r + r' + 90^\circ = 180^\circ \Rightarrow r' = 90 - r = 90 - i$$

$$n_1 \sin i = n_2 \sin r' = n_2 \sin(90 - i)$$

$$n_1 \sin i = n_2 \cos i \Rightarrow \tan i = \frac{n_2}{n_1}$$

$$\text{Now } \sin C = \frac{n_2}{n_1} = \tan i$$

$$\Rightarrow C = \sin^{-1}(\tan i) = \sin^{-1}(\tan r)$$

Q4

$$\text{At minimum deviation } r_1 = r_2 = \frac{A}{2}$$

$$\text{Also given } i = 2r_1 = A$$

Hints and Solutions

MathonGo

Now $1. \sin i = \sqrt{3} \sin r_1$

$$1 \sin A = \sqrt{3} \sin \frac{A}{2}$$

$$\Rightarrow 2 \sin \frac{A}{2} \cos \frac{A}{2} = \sqrt{3} \sin \frac{A}{2}$$

$$\Rightarrow \cos \frac{A}{2} = \frac{\sqrt{3}}{2} \Rightarrow \frac{A}{2} = 30^\circ$$

$$\Rightarrow A = 60^\circ$$

Q5

$$\sin \theta_C = \frac{1}{\mu} = \frac{1}{2\mu_2} < \sin \theta_C$$

$$\sin \theta > \sin \theta_C$$

$$\theta > \theta_c$$

Total internal reflection will happen

Q6

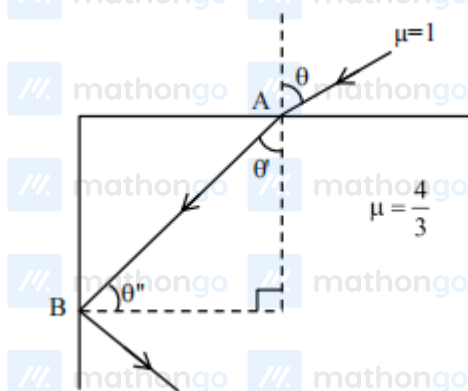
$$\mu = \frac{\sin\left(\frac{A+\delta_{\min}}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{\sin\left(\frac{A+A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{\sin A}{\sin \frac{A}{2}} = 2 \cos \frac{A}{2}$$

$$A = 2 \cos^{-1}\left(\frac{\mu}{2}\right)$$

Q7



At maximum angle θ ray at point B goes in grazing emergence, at all less values of θ , TIR occurs. At point B

Hints and Solutions

MathonGo

$$\frac{4}{3} \times \sin \theta'' = 1 \times \sin 90^\circ$$

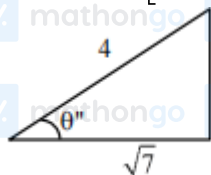
$$\theta'' = \sin^{-1} \left(\frac{3}{4} \right)$$

$$\theta' = \left(\frac{\pi}{2} - \theta'' \right)$$

At point A $1 \times \sin \theta = \frac{4}{3} \times \sin \theta'$

$$\sin \theta = \frac{4}{3} \times \sin \left(\frac{\pi}{2} - \theta'' \right)$$

$$\sin \theta = \frac{4}{3} \cos \left[\cos^{-1} \frac{\sqrt{7}}{4} \right]$$



$$\sin \theta = \frac{4}{3} \times \frac{\sqrt{7}}{4}$$

$$\theta = \sin^{-1} \left(\frac{\sqrt{7}}{3} \right)$$

Q8

For no bending, $n_1 = n_2$

$$1.2 + \frac{10.8 \times 10^{-14}}{\lambda^2} = 1.45 + \frac{1.8 \times 10^{-4}}{\lambda^2}$$

On solving,

$$9 \times 10^{-14} = 25\lambda^2$$

$$\lambda = 6 \times 10^{-7}$$

$$\lambda = 600 \text{ nm}$$

Q9

Standard graph between angle of deviation and incident angle.